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No : _____
Date : _____

1. Diberikan sebuah citra berukuran 8×8 piksel

3	0	3	2	0	5	0	3
1	6	1	9	2	3	5	5
5	2	2	4	6	2	6	0
6	2	4	2	3	0	1	7
0	3	2	0	1	6	6	2
5	4	2	5	9	2	0	3
2	6	6	7	7	7	7	6
1	2	4	0	2	0	2	1

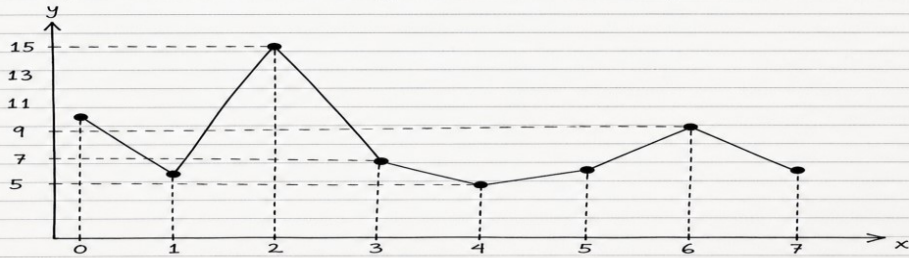
A. HISTOGRAM!

Langkah 1: Buat tabel frekuensi dari kemunculan setiap

Warna (x):	0	1	2	3	4	5	6	7
Frekuensi (y):	10	6	15	7	5	6	9	6

(Nilai intensitas yang sering muncul (y))

Langkah 2: Gambarkan histogram intensitas warnanya



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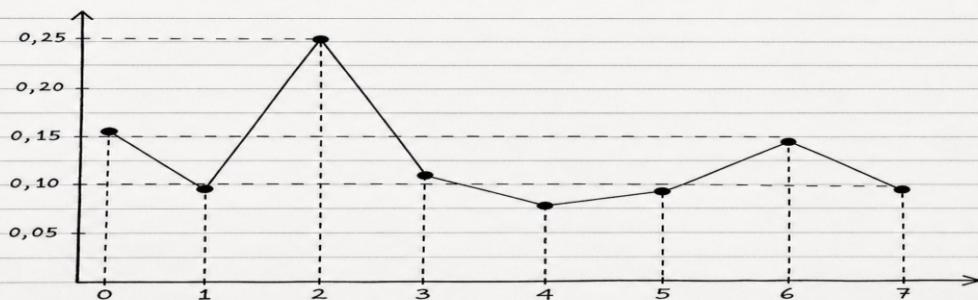
Titik kartesian : $(0, 10), (1, 6), (2, 15), (3, 7), (4, 5), (5, 6), (6, 9), (7, 6)$

B. Normalisasi Histogram

- Jumlah seluruh piksel = $8 \times 8 = 64$ piksel
- Rumus normalisasi = g / N
- Contoh : $10/64 = 0,15625$; $15/64 = 0,234375$; $9/64 = 0,140625$

Warna x	0	1	2	3	4	5	6	7
Jumlah y	10	6	15	7	5	6	9	6
g / N	0,15625	0,09375	0,234375	0,109375	0,078125	0,09375	0,140625	0,09375

- Gambarkan histogram atau kurva setelah dinormalisasi



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Commands + Code + Text ▶ Run all

[1]
✓ 1m
import cv2
import matplotlib.pyplot as plt
from google.colab import files

# Upload gambar dari laptop
uploaded = files.upload()

BOLA.avif
BOLA.avif(image/avif) - 31644 bytes, last modified: n/a - 100% done
Saving BOLA.avif to BOLA.avif

[2]
✓ 0s
# Mengambil nama file gambar yang diupload
image_path = list(uploaded.keys())[0]

# Membaca gambar dalam mode warna asli BGR
image = cv2.imread(image_path)

[3]
✓ 1s
▶ # Mengubah BGR menjadi RGB untuk matplotlib
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)

# Memisahkan kanal warna RGB
red_channel = image_rgb[:, :, 0]
green_channel = image_rgb[:, :, 1]
blue_channel = image_rgb[:, :, 2]

# Membuat histogram untuk masing-masing kanal warna
hist_red = cv2.calcHist([red_channel], [0], None, [256], [0, 256])
hist_green = cv2.calcHist([green_channel], [0], None, [256], [0, 256])
hist_blue = cv2.calcHist([blue_channel], [0], None, [256], [0, 256])
```

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[1]
✓ 1s
▶ # Menampilkan gambar asli dan histogram masing-masing kanal warna
plt.figure(figsize=(8, 8))

# Menampilkan gambar asli
plt.subplot(2, 2, 1)
plt.imshow(image_rgb)
plt.title('Citra Asli')
plt.axis('off')

# Menampilkan histogram merah
plt.subplot(2, 2, 2)
plt.plot(hist_red, color='r')
plt.title('Red Channel Histogram')
plt.xlabel('Intensitas Piksel')
plt.ylabel('Jumlah Piksel')

# Menampilkan histogram hijau
plt.subplot(2, 2, 3)
plt.plot(hist_green, color='g')
plt.title('Green Channel Histogram')
plt.xlabel('Intensitas Piksel')
plt.ylabel('Jumlah Piksel')

# Menampilkan histogram biru
plt.subplot(2, 2, 4)
plt.plot(hist_blue, color='b')
plt.title('Blue Channel Histogram')
plt.xlabel('Intensitas Piksel')
plt.ylabel('Jumlah Piksel')

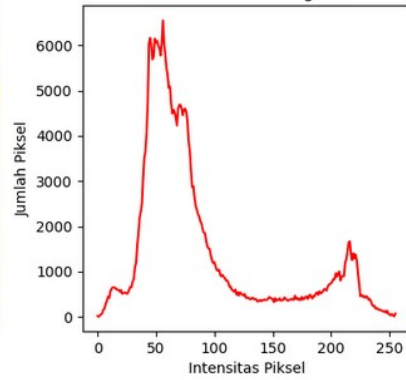
plt.tight_layout()
plt.show()
```

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plt.show()
```

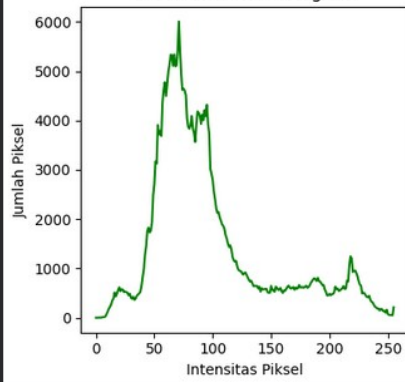
Citra Asli



Red Channel Histogram



Green Channel Histogram



Blue Channel Histogram

